

Episcleral plaque ^{125}I radiotherapy with episcleral LCF hyperthermia: A prospective randomized trial

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ABSTRACT

PURPOSE: The purpose of this study was to search for an optimal radiation dose in the treatment of patients with uveal melanoma using ^{125}I episcleral plaque radiotherapy (EPRT) and episcleral hyperthermia (HT).

METHODS AND MATERIALS: From 1991–1998, 35 patients with uveal melanoma were enrolled in a phase II prospective randomized trial of ^{125}I EPRT combined with episcleral HT. Two groups were closely matched for pre-treatment patient and tumor characteristics. Group 1: N = 16, and Group 2: N = 19. The median dose to the tumor apex for Group 1 was 80.0 Gy and 60.8 Gy for Group 2. Episcleral HT was given once for 45 min immediately prior to EPRT with a median temperature of 44°C for both groups. The median follow-up was 5.5 years for Group 1 and 5.3 years for Group 2.

RESULTS: The median tumor height decreased 1.7 mm for patients of both groups. The 5- and 8-year probability of local recurrence was 33% for Group 1, and 25% for Group 2, $p = 0.73$. The 5-year probability of DFS was 54% for Group 1 and 67% for Group 2, $p = 0.51$. The 5- and 8-year overall survival was 68% and 34%, respectively, for Group 1, and 83% and 50%, respectively, for Group 2, $p = 0.60$. The rate of distant metastasis at 5- and 8-years for Group 1 was 29% and 62%, respectively, and 17% and 17%, respectively, for Group 2, $p = 0.18$. The incidence of enucleation was 4 (25%) in Group 1 vs. 4 (22%) in Group 2. The incidence of late complications was similar in either treatment group. The ambulatory visual acuity ($>5/200$) at last follow-up was slightly better in Group 2 (80%) than Group 1 (64%).

CONCLUSIONS: Treatment outcomes were similar despite a 25% difference in radiation dose. In view of these findings and in an attempt to reduce the incidence of late treatment toxicity a still lower radiation dose in combination with HT needs to be studied. The reported outcomes need to be evaluated with caution due to the small number of patients in this study. © 2003 American Brachytherapy Society. All rights reserved.

Keywords:

Uveal; Melanoma; Plaque; Thermoradiotherapy; Episcleral

Introduction

There are several treatment options available for the management of patients with uveal melanoma. Enucleation and

episcleral plaque radiotherapy (EPRT) are the most commonly used treatment modalities (1, 2). In the past 20 years, EPRT has established itself as a well-recognized and effective treatment for T1, T2, and selected T3 lesions (3–5). Results of EPRT have been excellent with consistently reported 5-year actuarial survival of more than 75% (6–9). The most conclusive evidence for the effectiveness of EPRT for uveal melanoma when compared with enucleation was provided by a multi-institutional, prospective randomized trial of 1317 patients conducted by the Cooperative Ocular Melanoma Study (COMS) (10, 11). This important report demonstrated a similar 5- and 10-year overall survival for

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